

She's a Replicant, Blade Runner, 1982, Ridley Scott
https://www.youtube.com/watch?v=yWPyRSURYFQ

Outline: W02C-W07B

I) Task: Classification

II) Supervision: Fully Supervised

III) Technique: k-Nearest Neighbor

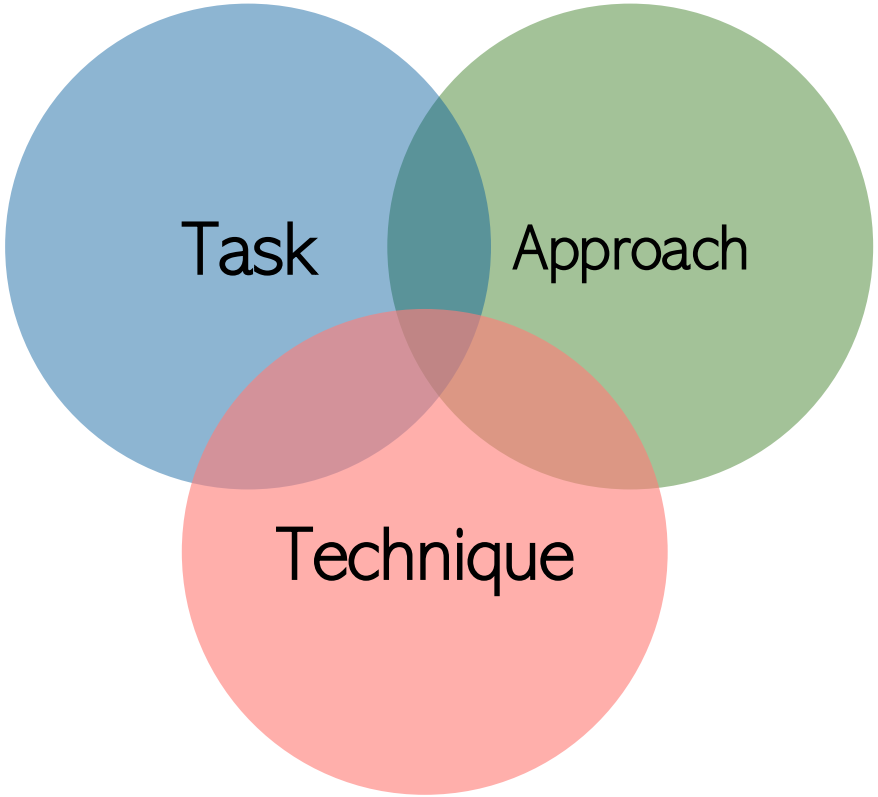
IIII) Evaluation

Metrics, Methodology

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Outline: W07C: New Technique: Logistic Regression (LR)

Classification
Regression
Clustering
Ranking (Recommendation)
Vector Representation and Dimensionality Reduction
Language Modeling



Full/Self/Semi-**Supervised Learning**
Unsupervised Learning
Reinforcement Learning

k-Nearest Neighbor, Naive Bayes, **Logistic Regression**
Artificial Neural Networks, Graph Neural Network
Support Vector Machines
Principal Component Analysis
K-Means

Outline: W07C

I) Task: Classification

II) Supervision: Fully Supervised

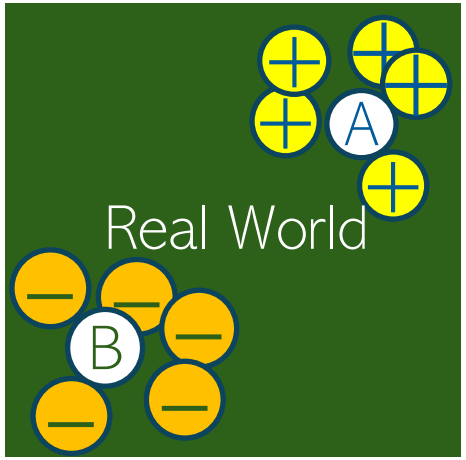
III) Technique: Logistic Regression → Neural Networks

IIII) Evaluation

Metrics (Few New Metrics), Methodology

kNN is a Locally Affinitive Learner

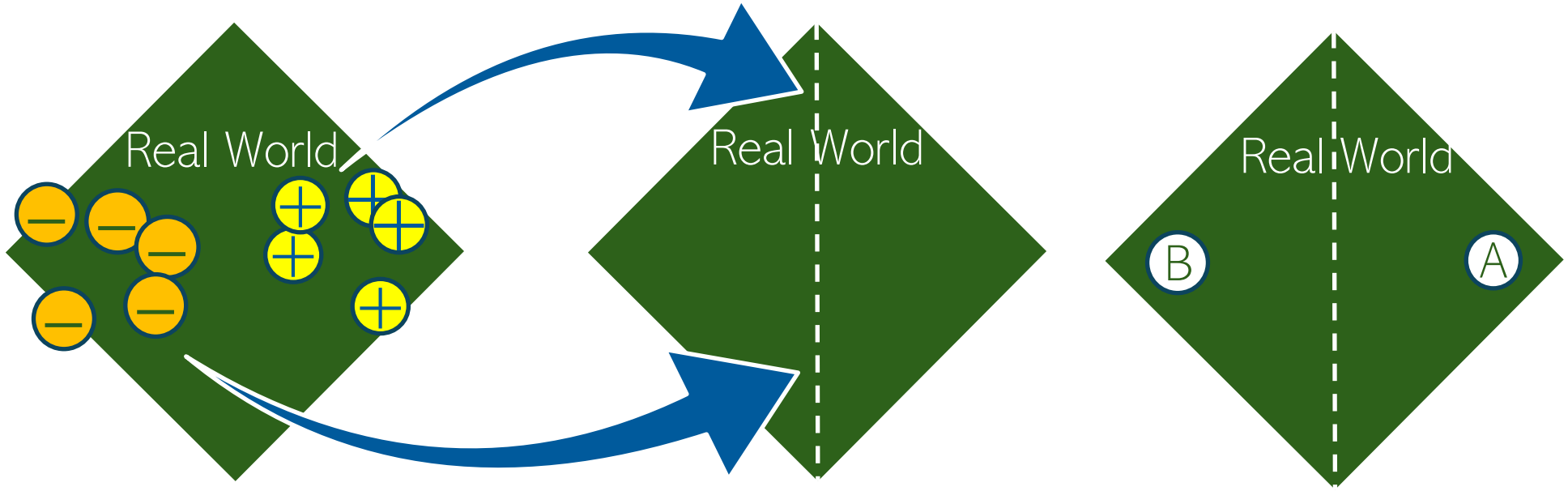
2-NN(A) = 1
2-NN(B) = 0



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A Technique based on a Global Discriminator

Summarizing the Entire World into a Discrimination Boundary



f(A) = Far Right = Very Pos
f(B) = Far Left = Very Neg

Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0$

The universe can be summarized (learned) into a number:

- 1 parameter (memory cell, weight)

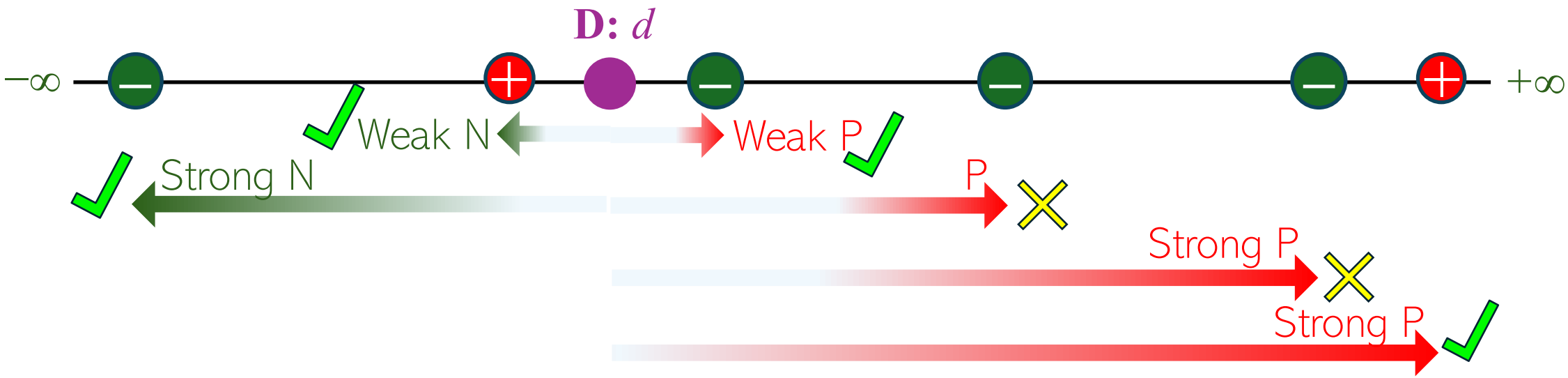


$$f(\mathbf{o}) = d(\mathbf{o}, \mathbf{D})$$
$$= o_i - d$$
$$= \begin{cases} > 0\text{: larger positive, stronger discrimination for being positive} \\ < 0\text{: smaller negative, stronger discrimination for being negative} \end{cases}$$



Discriminator in Euclidian Space

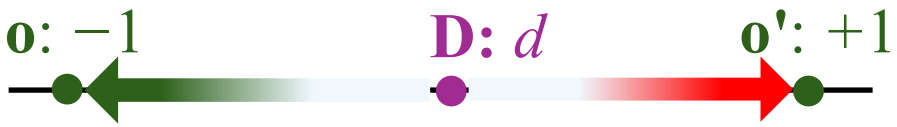
1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0$



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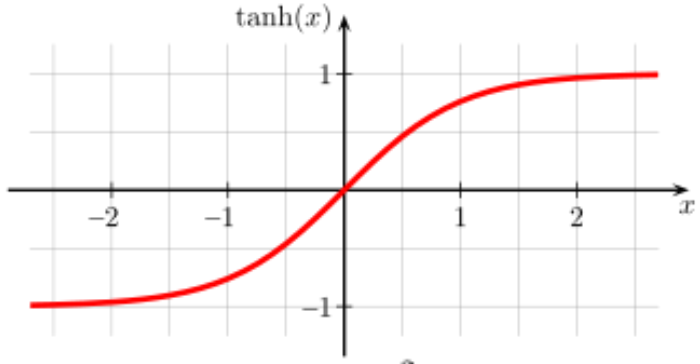
Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \longrightarrow \mathbb{R}^0 \longrightarrow \mathbb{R}^{[-1,+1]}$



$$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$$

$$= tanh(o_i - d)$$



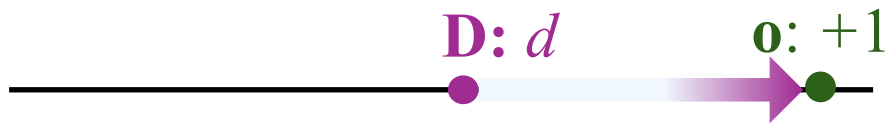
$$\tanh(z) = \frac{\mathrm{e}^{z} - \mathrm{e}^{-z}}{\mathrm{e}^{z} + \mathrm{e}^{-z}} = \frac{\mathrm{e}^{2z} - 1}{\mathrm{e}^{2z} + 1} = \frac{1 - \mathrm{e}^{-2z}}{1 + \mathrm{e}^{-2z}}.$$

$$= \begin{cases} > 0\text{: being positive, the more toward } +1\text{, stronger positive} \\ < 0\text{: being negative, the more toward } -1\text{, stronger negative} \end{cases}$$

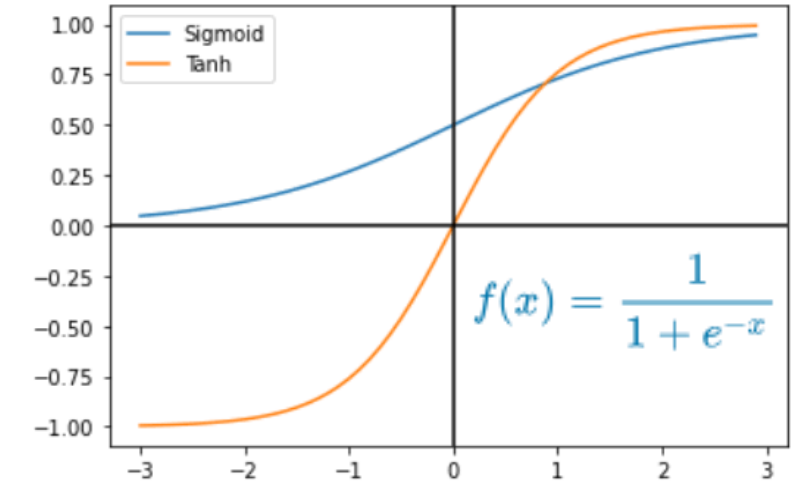
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Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]}$

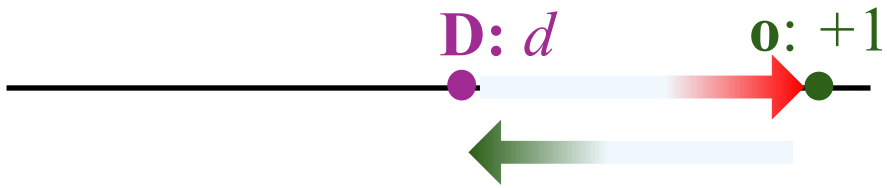


$$
\begin{aligned}
f(\mathbf{o}) &= d(\mathbf{o}, \mathrm{D}) \\
&= logistic\ curve(o_i - d) \\
&= logistic\ regression(o_i - d) \\
&= sigmoid(o_i - d) \\
&= \begin{cases} \text{The more toward } +1\text{, stronger positive} \\ \text{The less toward } +1\text{, weaker positive} \end{cases}
\end{aligned}
$$



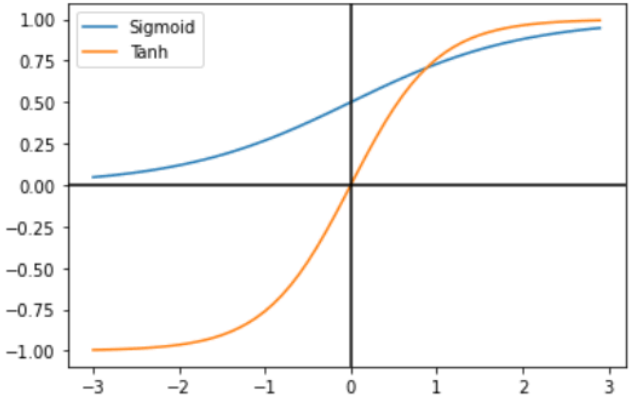
Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \longrightarrow \mathbb{R}^0 \longrightarrow \mathbb{R}^{[0,+1]} \longrightarrow$ Probability

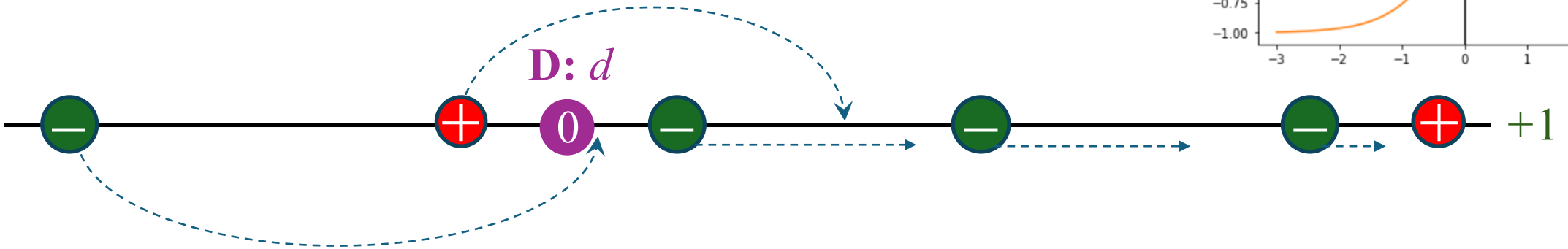


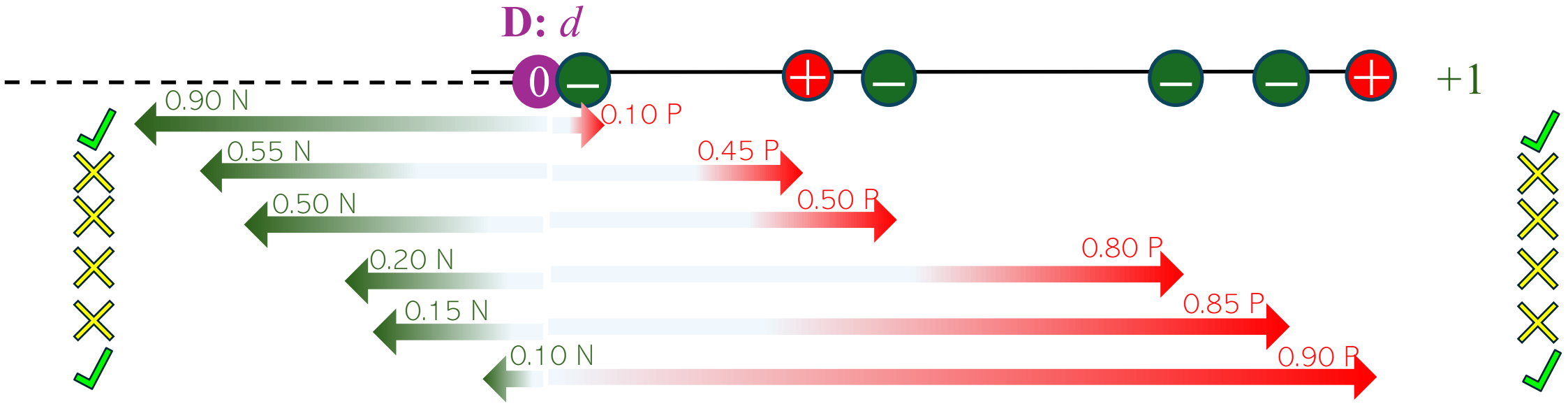
$$
\begin{aligned}
f(\mathbf{o}) &= d(\mathbf{o}, \mathrm{D}) \\
&= logistic\ curve(o_i - d) \\
&= logistic\ regression(o_i - d) \\
&= sigmoid(o_i - d) \\
&= \begin{cases} \text{The more toward } +1\text{, stronger probability for being positive} \\ 1 - \text{probability for being positive} = 1 - sigmoid(o_i - d). \end{cases}
\end{aligned}
$$

Discriminator in Euclidian Space



1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability





Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1$

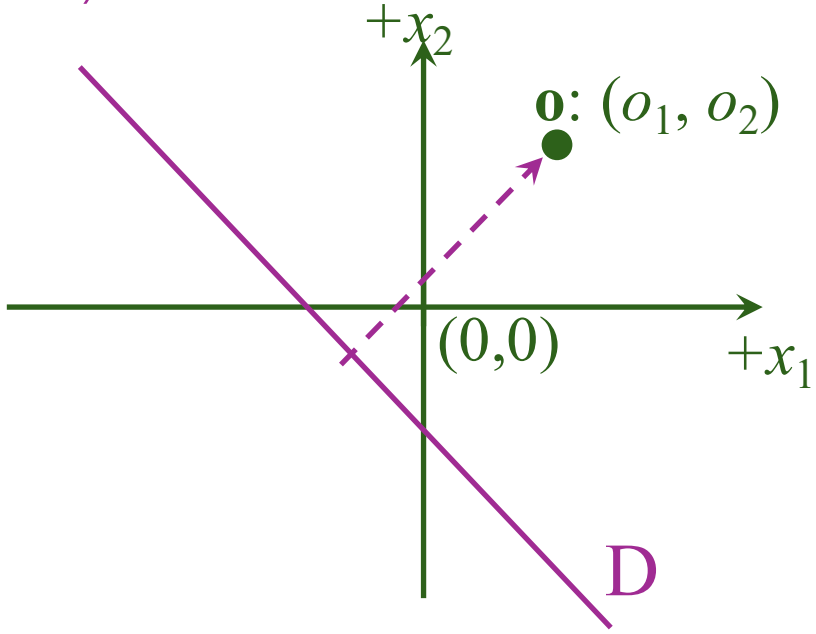
The universe can be summarized (learned) into a line:

- 3 parameters (coefficients, weights, memory cells)

$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$?

$$\mathrm{D}: \begin{bmatrix} x_1 & x_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = w_1 x_1 + w_2 x_2 + w_3 = 0$$

$$\mathrm{D}: x_2 = \frac{-1}{w_2}(w_1 x_1 + w_3) = -\frac{w_1}{w_2} x_1 + \frac{w_3}{w_2}$$



https://www.desmos.com/calculator

Discriminator: Euclidian Space

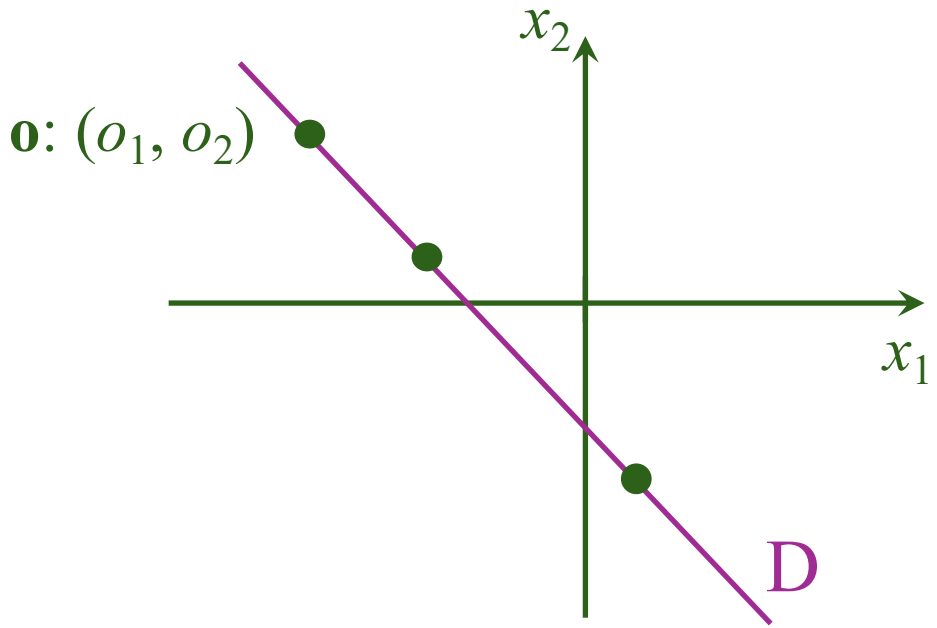
2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1$

From math, the line equation is true for all points on the line.

We can think of it as distance measurement: all points on the line has distance $\mathbf{0}$.

$$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D}) \ ?$$

$$\mathrm{D}: \begin{bmatrix} o_1 & o_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = w_1 o_1 + w_2 o_2 + w_3 = 0$$



https://www.desmos.com/calculator

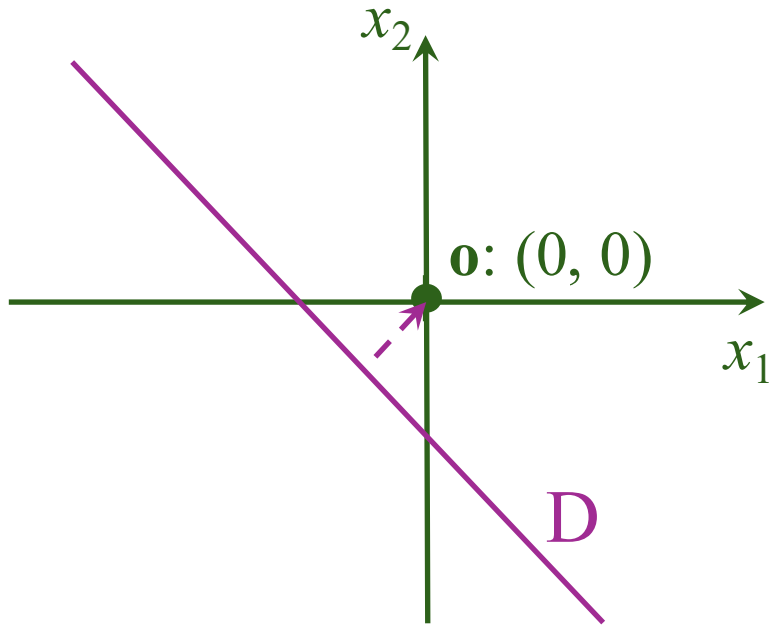
Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1$

What happens for distance of a point outside the line?

$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$?

$$\mathrm{D}: [0 \quad 0 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = w_1 0 + w_2 0 + w_3 \neq 0$$



https://www.desmos.com/calculator

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Discriminator: Euclidian Space

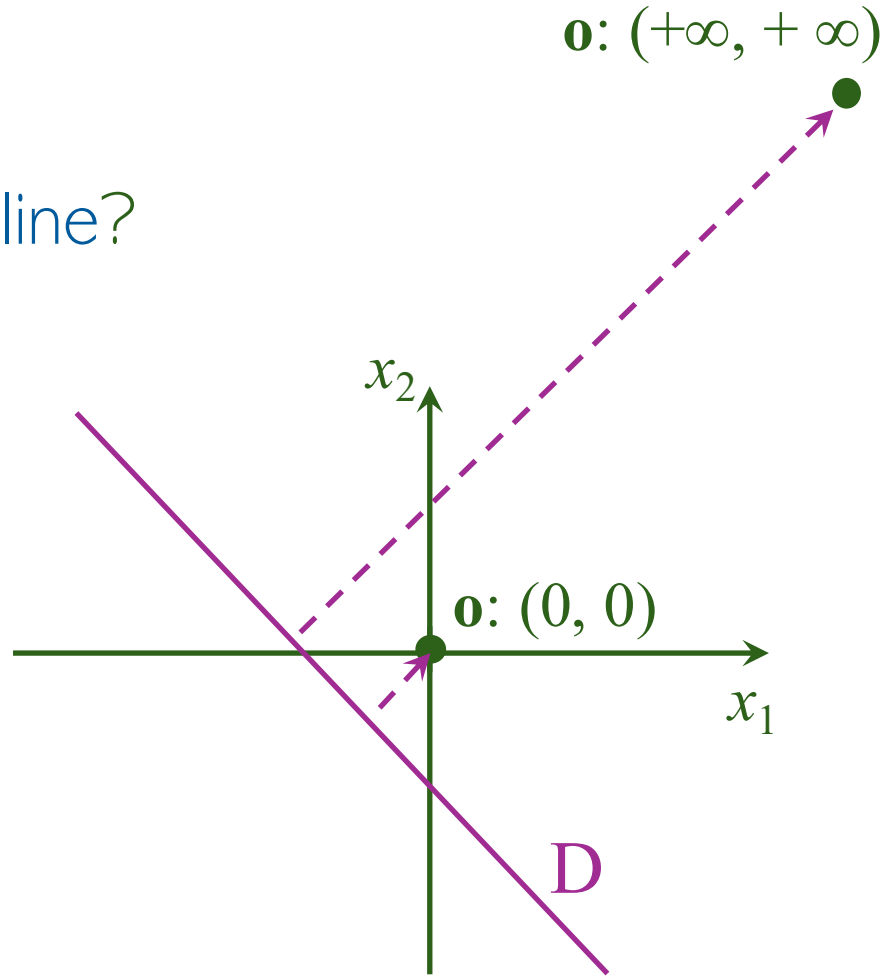
2D Euclidian Distance $\mathbb{R}^2 \longrightarrow \mathbb{R}^1$

What happens for distance of a point outside the line?

$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$?

$$\mathrm{D}: [0 \quad 0 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = w_1 0 + w_2 0 + w_3 > 0$$

$$\mathrm{D}: [+\infty \quad +\infty \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = w_1 \infty + w_2 \infty + w_3 \longrightarrow +\infty$$



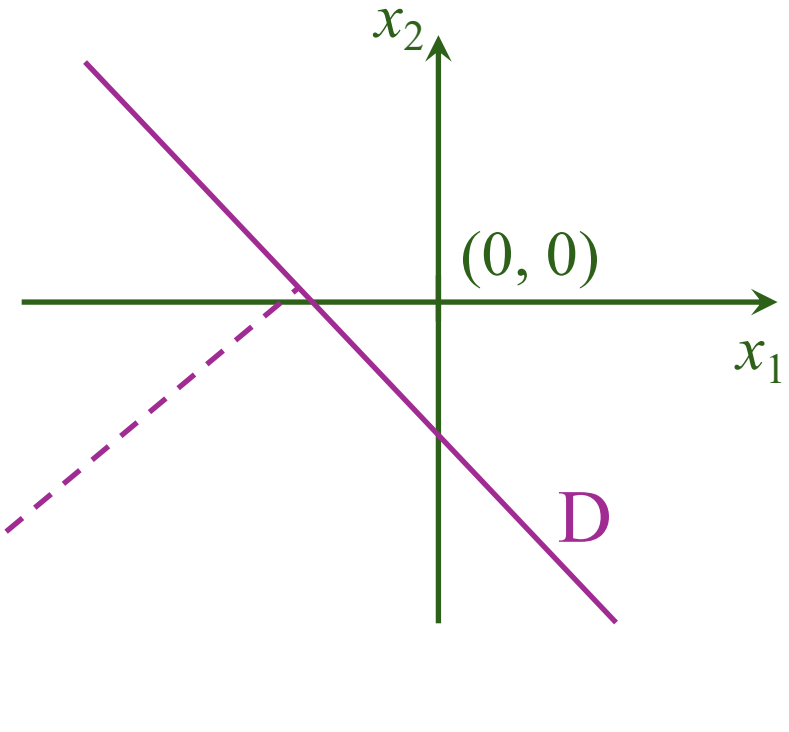
Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1$

What happens for distance of a point outside the line?

$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$?

$$\mathrm{D}: \begin{bmatrix} -\infty & -\infty & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = -w_1\infty - w_2\infty + w_3 \rightarrow -\infty$$



Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^2 \longrightarrow \mathbb{R}^1$

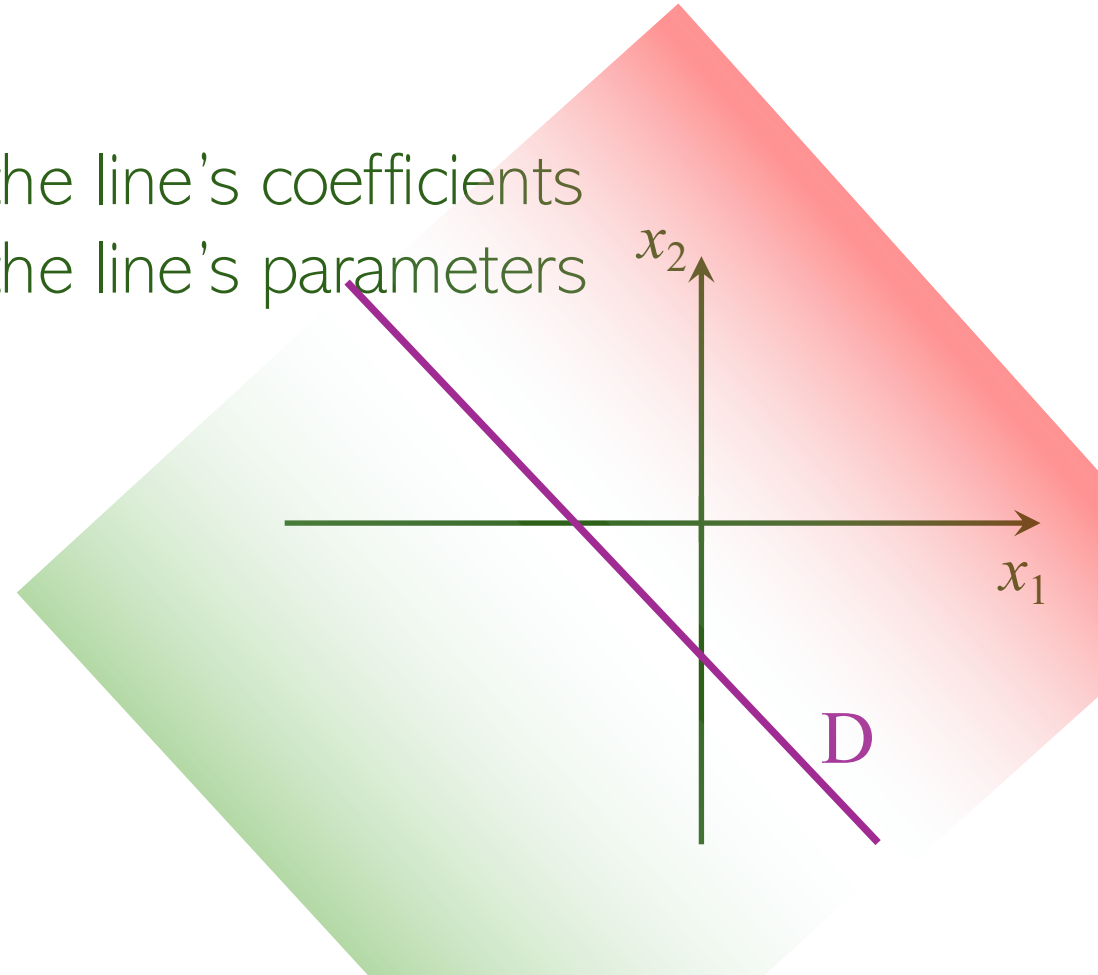
$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$

$\approx$ multiplication of the point in $\mathbb{R}^{2+1}$ by the line's coefficients

$\approx$ multiplication of the point in $\mathbb{R}^{2+1}$ by the line's parameters

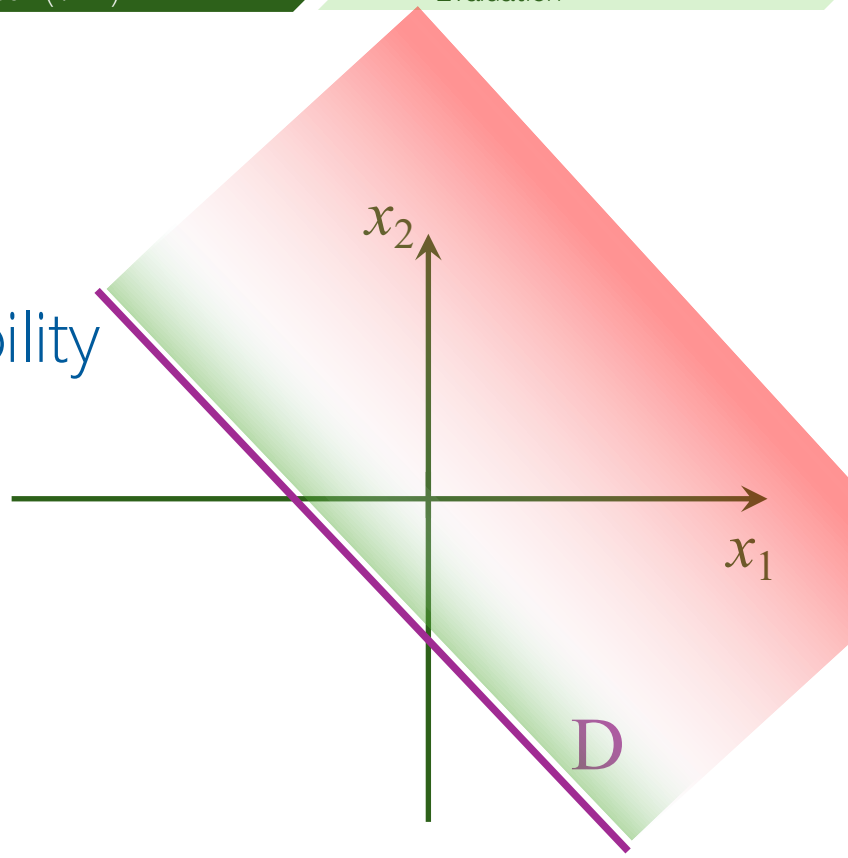
$$d\colon \begin{bmatrix} x_1 & x_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} > 0 \text{ if above the line}$$

$$d\colon \begin{bmatrix} x_1 & x_2 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} < 0 \text{ if below the line}$$



Discriminator: Euclidian Space

2D Euclidian Distance $\mathbb{R}^2 \rightarrow \mathbb{R}^1 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability



$$f(\mathbf{o}) = d(\mathbf{o}, \mathrm{D})$$

$$\approx sigmoid([x_1 \quad x_2 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix})$$

$$\approx \begin{cases} \text{The more toward } +1\text{, stronger probability for being positive} \\ 1 - \text{probability for being positive} = 1 - sigmoid([x_1 \quad x_2 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}) \end{cases}$$

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Discriminator: Euclidian Space

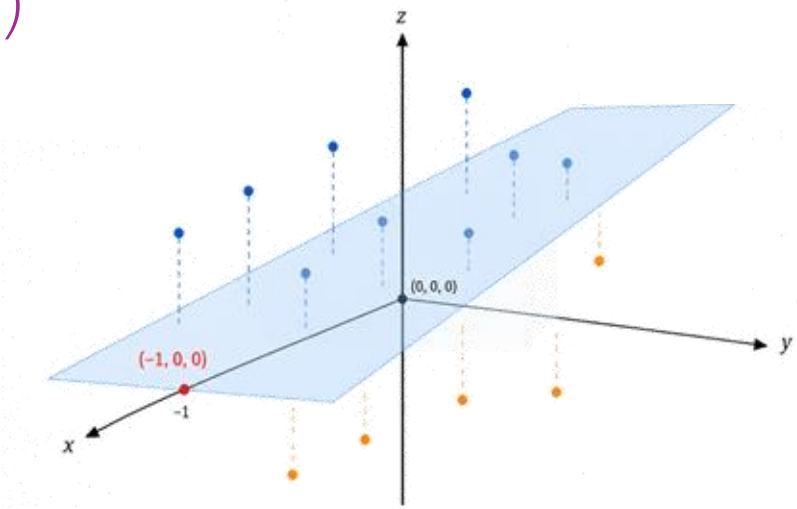
2D Euclidian Distance $\mathbb{R}^3 \longrightarrow \mathbb{R}^2$

The universe can be summarized (learned) into a plane:

- 4 parameters (coefficients, weights, memory cells)

$\longrightarrow \mathbb{R}^{[0,+1]} \longrightarrow$ Probability

$$f(\mathbf{o}) = d(\mathbf{o}, \mathbf{D}) \approx sigmoid([x_1 \quad x_2 \quad x_3 \quad 1] \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix})$$

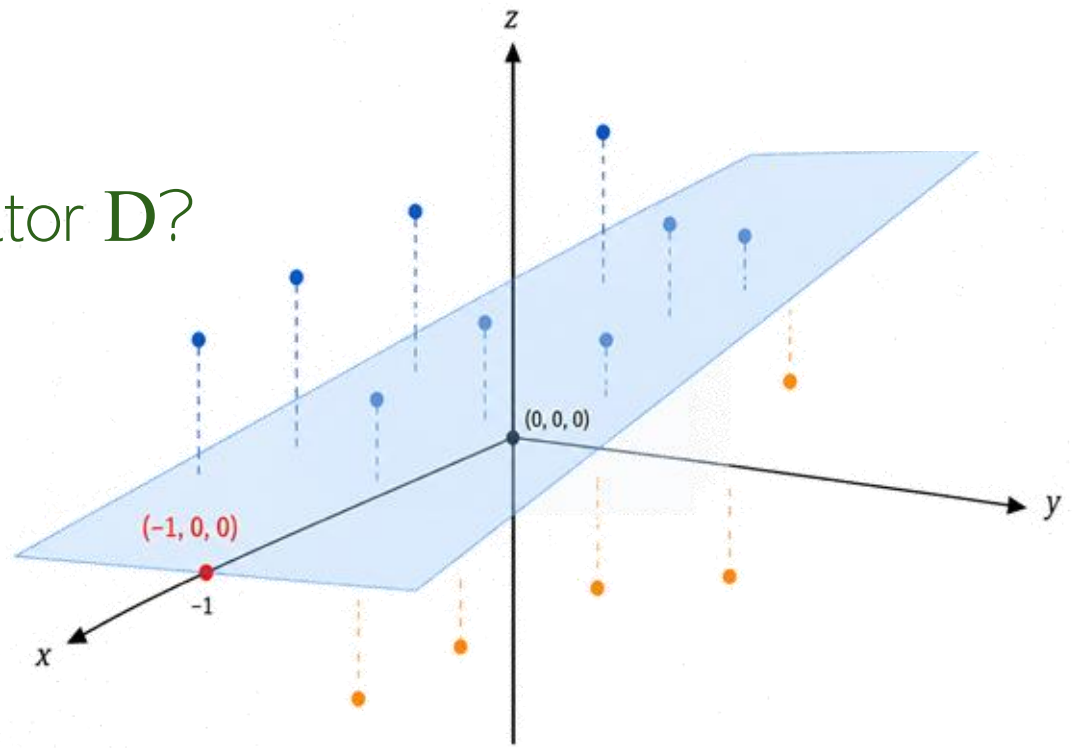


$$\approx \begin{cases} \text{The more toward } +1\text{, stronger } \textbf{probability} \text{ for being positive} \\ 1 - \text{probability for being positive} = 1 - sigmoid(\ldots) \end{cases}$$

Discriminator: Euclidian Space

What is, or who knows, the true discriminator $\mathbf{D}$?

- Point
- Line
- Plane
- Hyperplane



No one!

Aletheia, the goddess of truth
God, in the Abrahamic religions as the all-powerful, all-knowing, eternal deity, and single creator of the universe
Omniscient /äm'niSHənt/: omni=all, scientia= knowledge or to know
The Oracle of Delphi
Teacher
Supervisor
Human (Expert) Annotator
Majority Vote on Human (Expert) Annotators
LLM-as-a-Judge

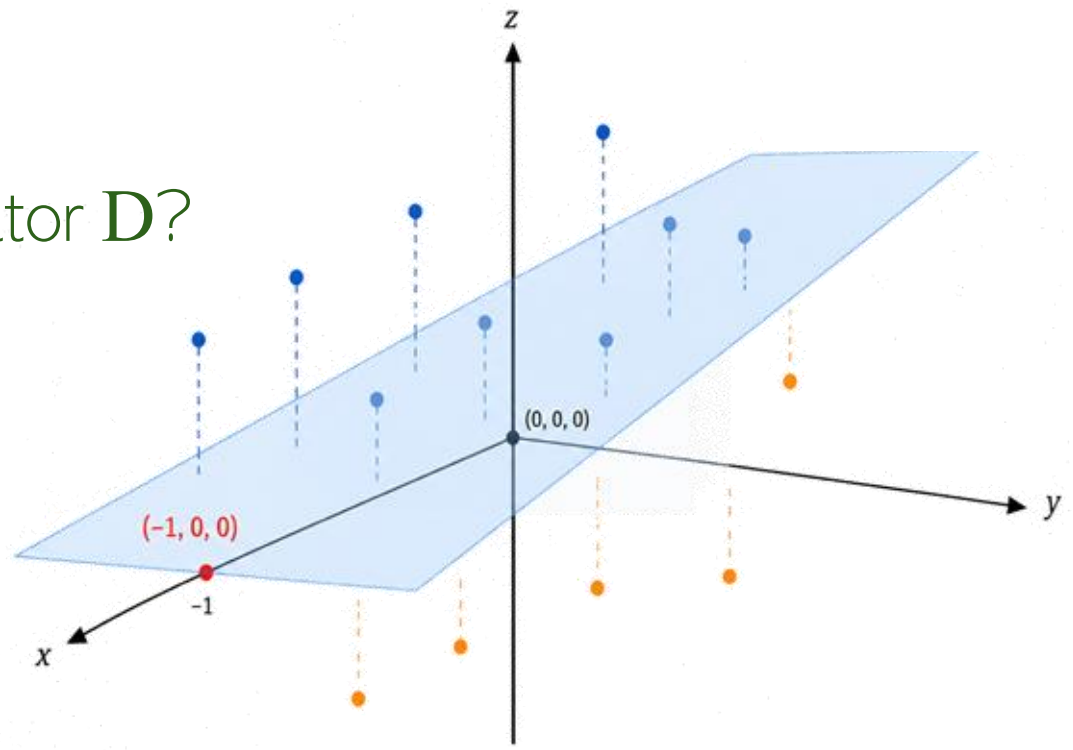




Discriminator: Euclidian Space

What is, or who knows, the true discriminator $\mathbf{D}$?

- Point
- Line
- Plane
- Hypeplane



We find it through an iterative process:

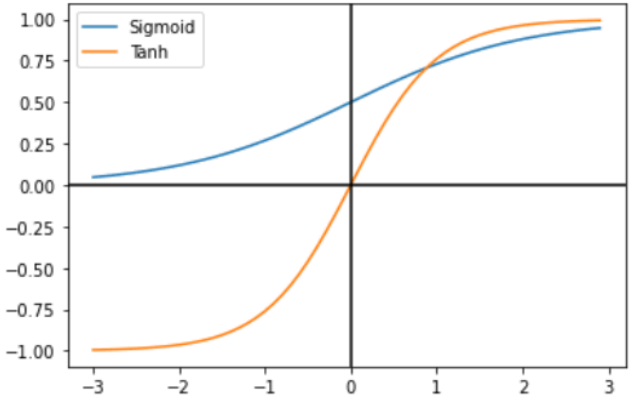
1. Initial random guess
2. Evaluate on the train set
3. Tune the discriminator's parameters toward *almost* overfitting
4. Back to (2)

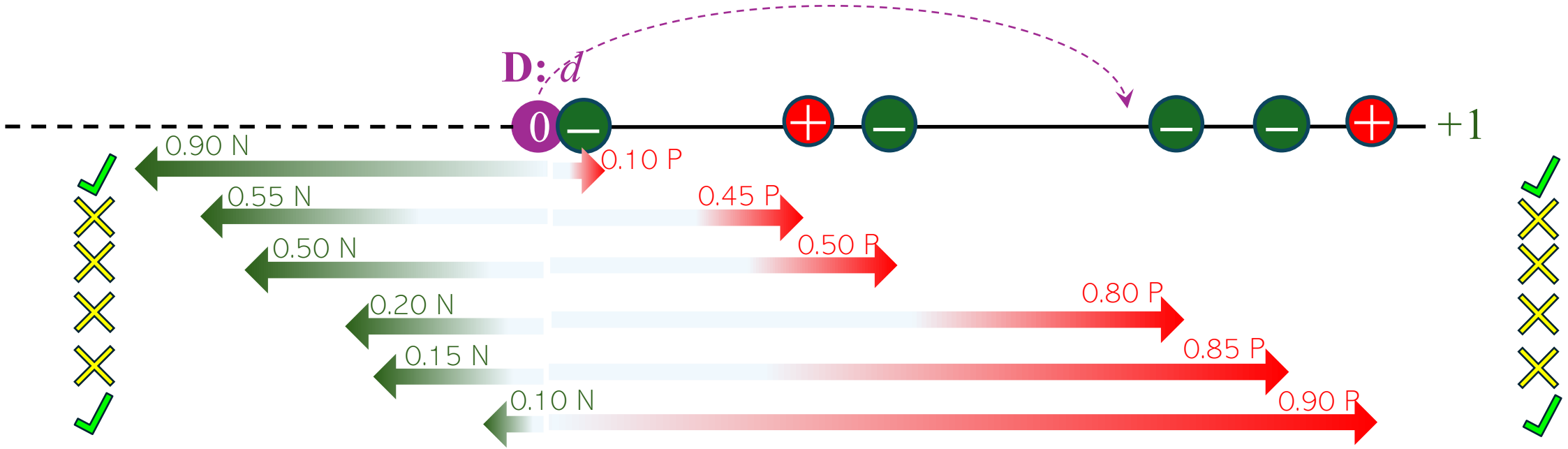




Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability





Discriminator in Euclidian Space

1D Euclidian Distance $\mathbb{R}^1 \rightarrow \mathbb{R}^0 \rightarrow \mathbb{R}^{[0,+1]} \rightarrow$ Probability

